

01-02B ON-BOARD DIAGNOSTIC [WL-C, WE-C]

ON-BOARD DIAGNOSTIC OUTLINE

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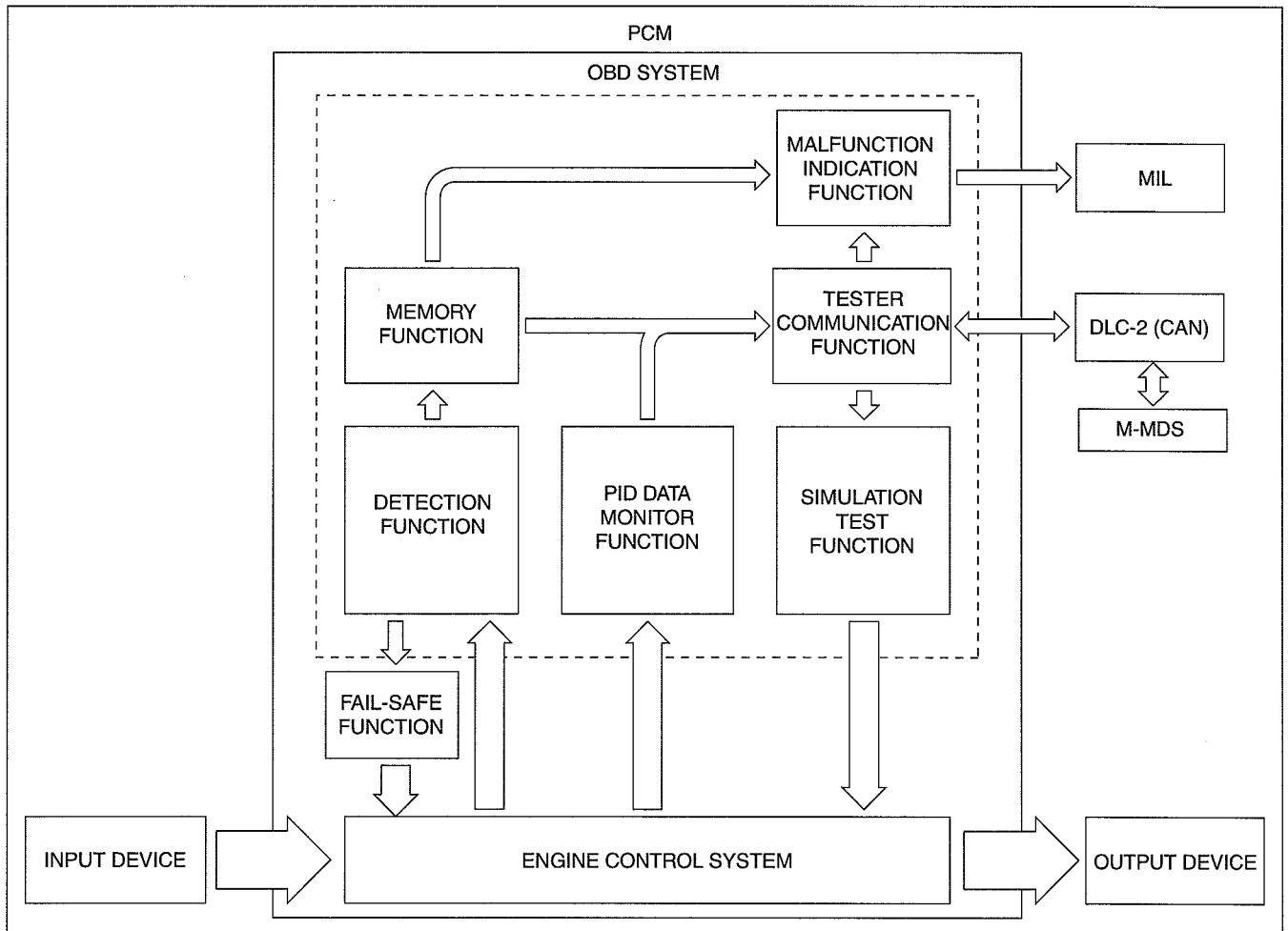
ON-BOARD DIAGNOSTIC OUTLINE [WL-C, WE-C]

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Features

Improved serviceability	• DTCs adopted • KOEO/KOER self-test function adopted • PID/DATA monitor function adopted • Simulation test function adopted
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Block Diagram



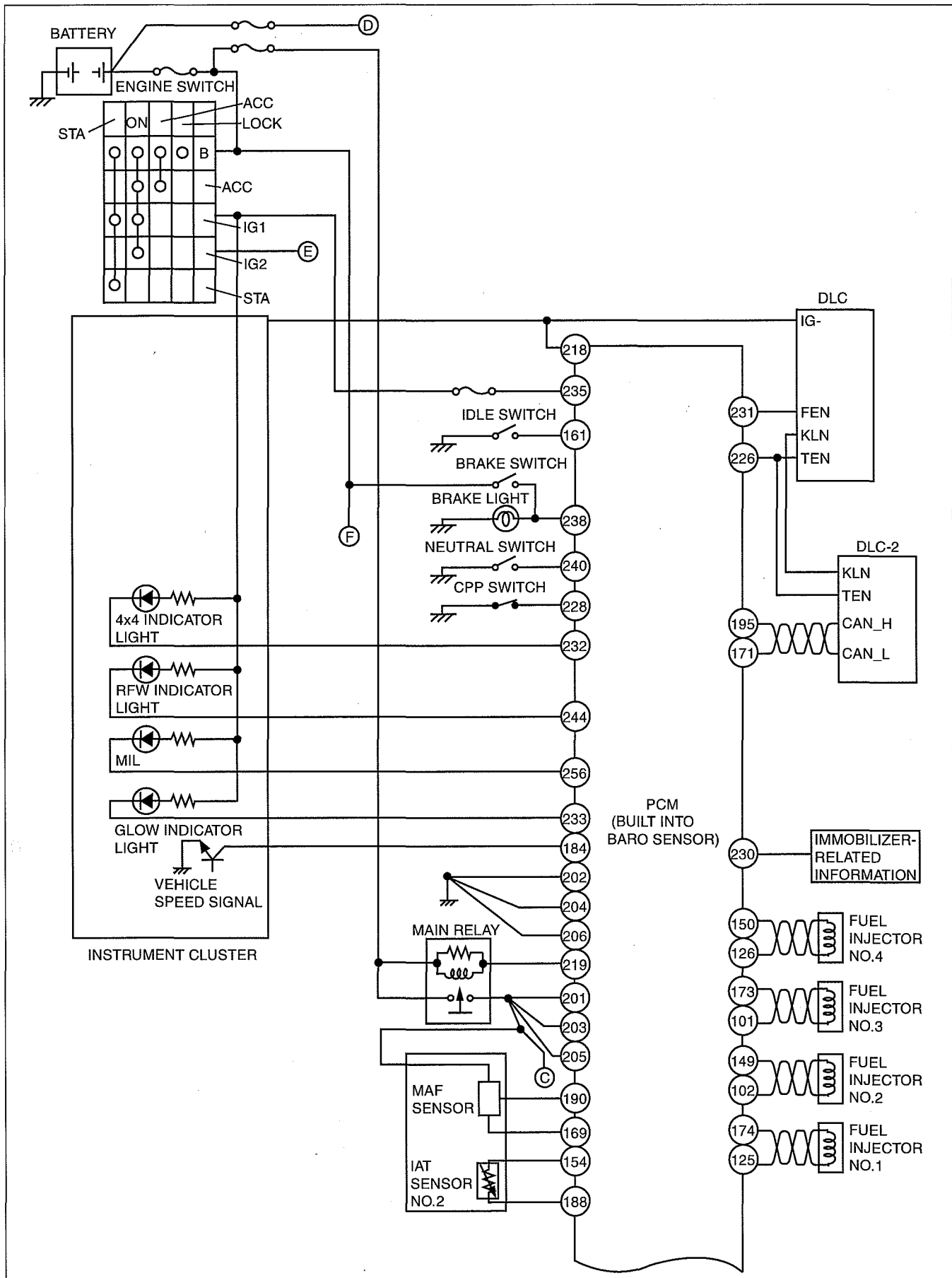
DBG102BTB001

ON-BOARD DIAGNOSTIC [WL-C, WE-C]

DIAGNOSTIC SYSTEM WIRING DIAGRAM [WL-C, WE-C]

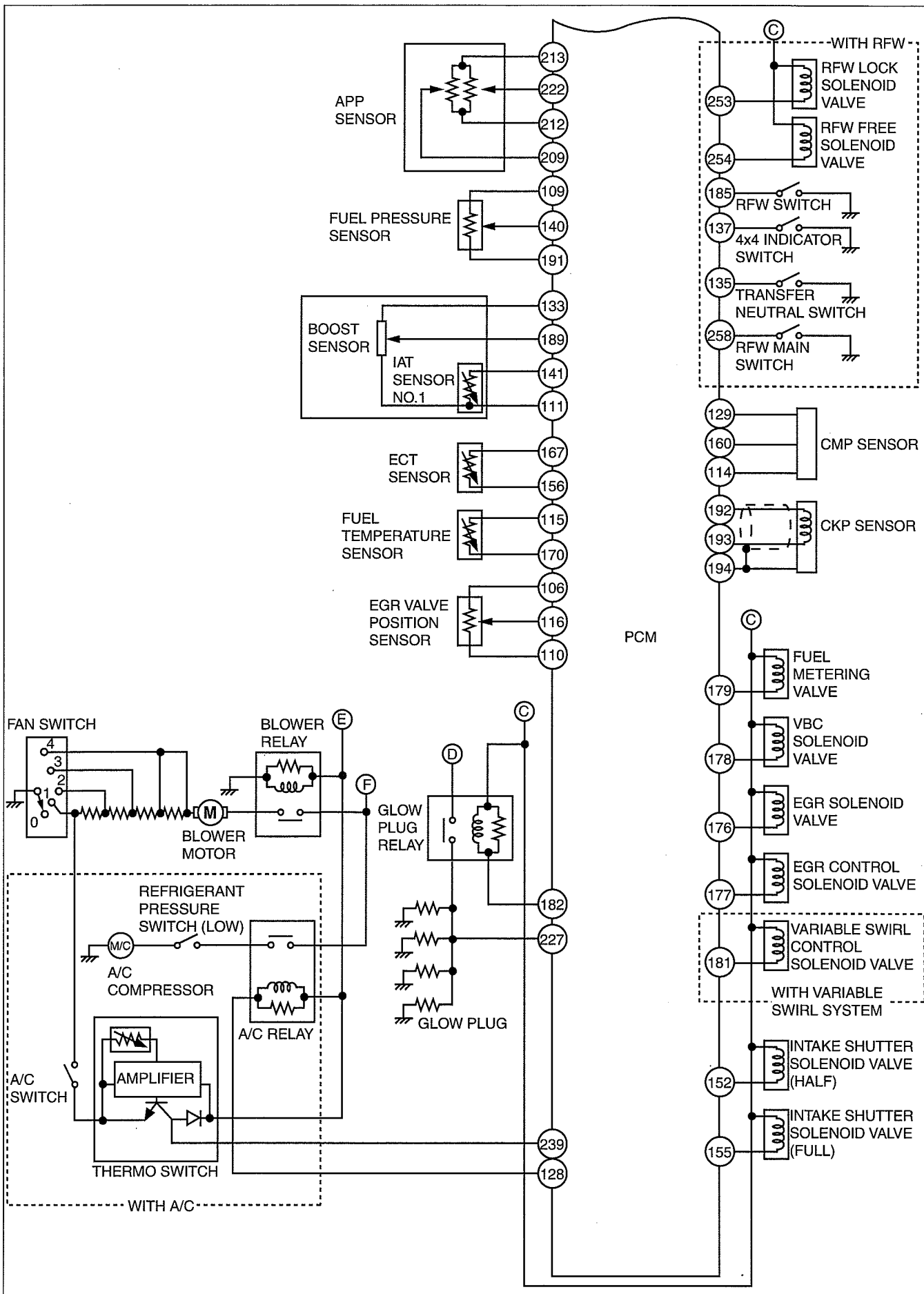
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With Immobilizer System



DBG102BWB01

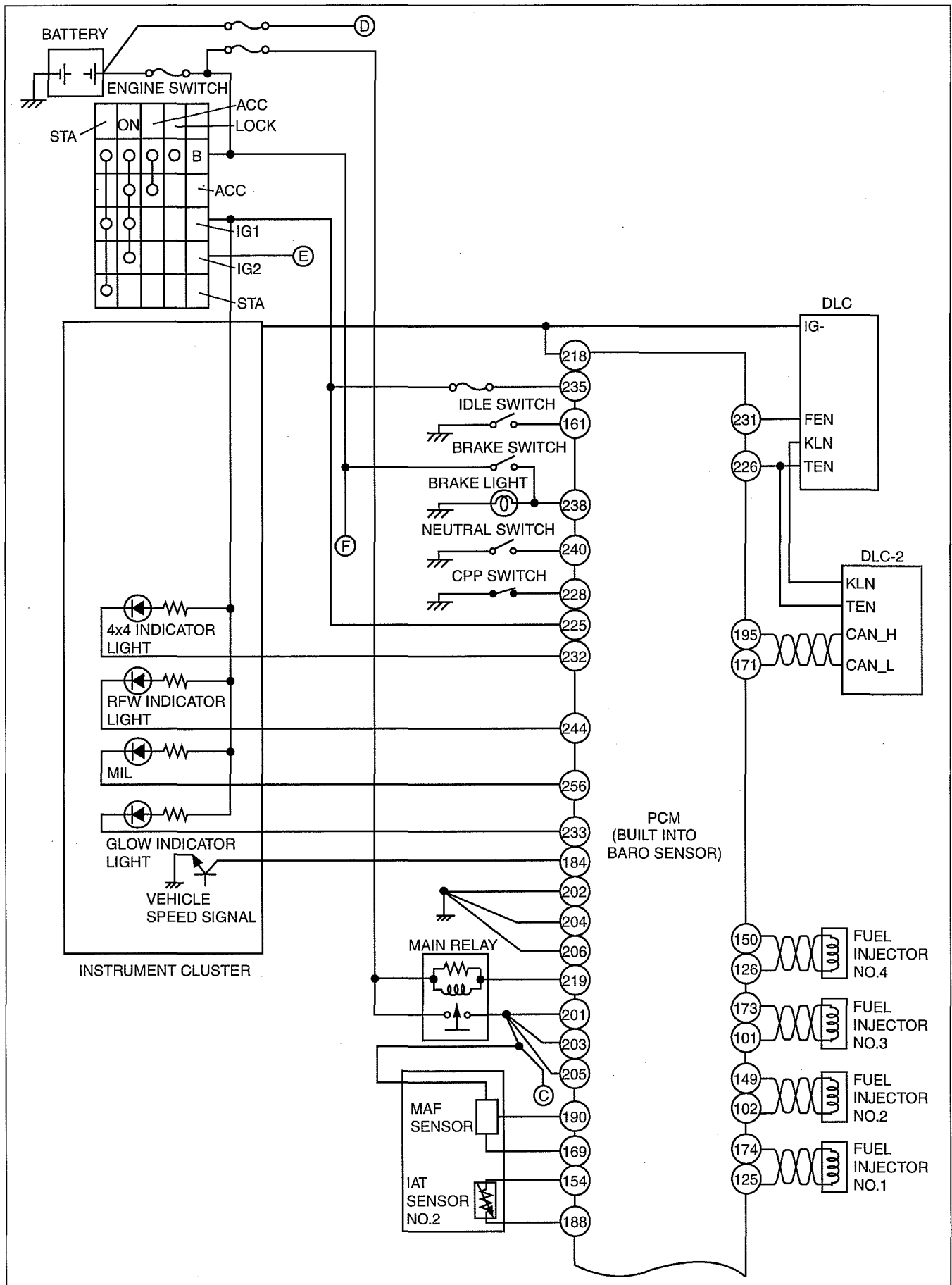
ON-BOARD DIAGNOSTIC [WL-C, WE-C]



DBG102BWBY02

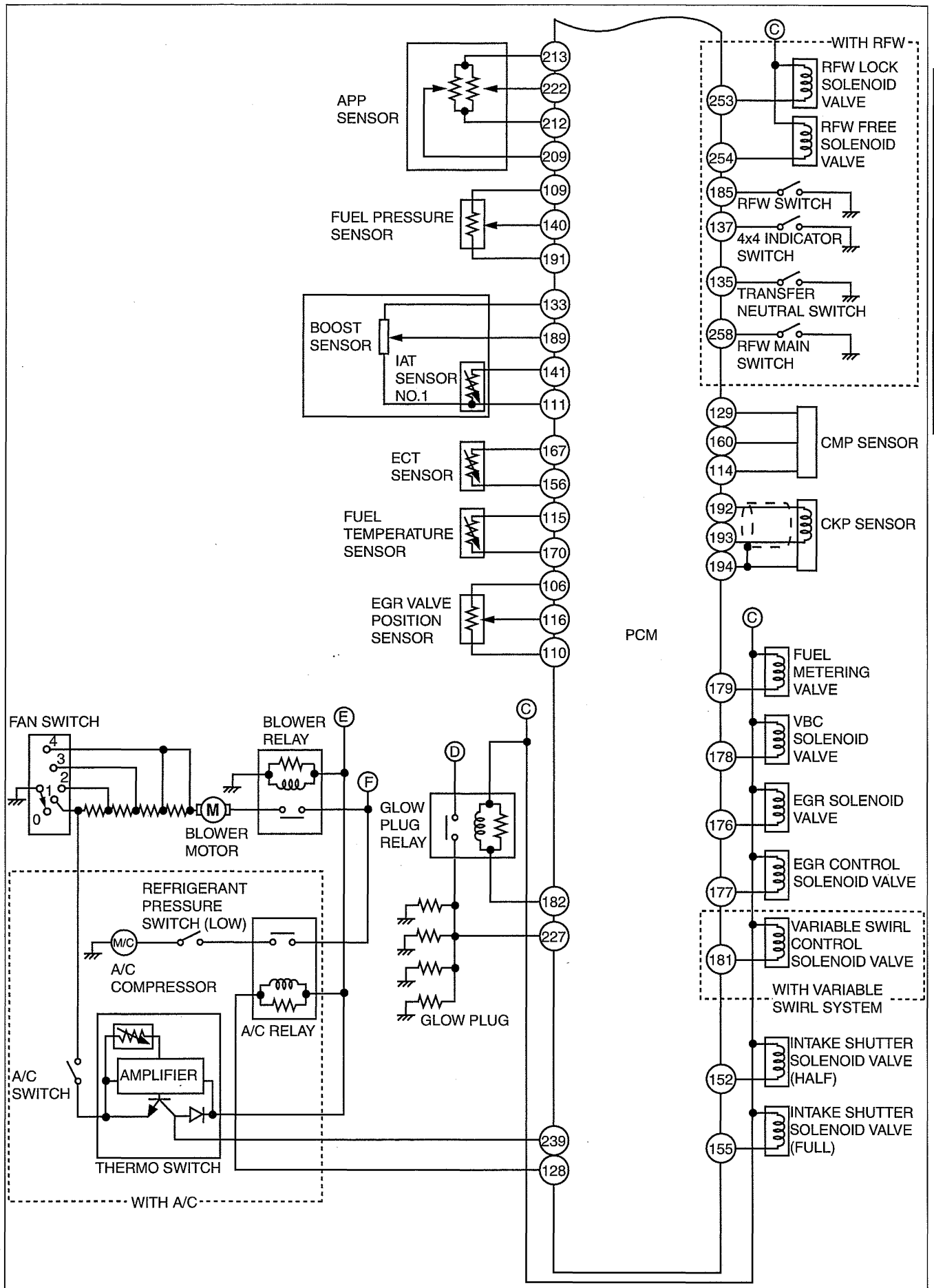
ON-BOARD DIAGNOSTIC [WL-C, WE-C]

Without Immobilizer System



DBG102BWBY03

ON-BOARD DIAGNOSTIC [WL-C, WE-C]



DBG102BWB02

ON-BOARD DIAGNOSTIC [WL-C, WE-C]

DTC TEST MODE [WL-C, WE-C]

dcf010200000115

Sending Diagnostic Data

PID data monitor

- The PID data monitor items are shown below.

PID data monitor table

—: Not applicable

Full names	Unit	
Monitor status since DTCs cleared	—	
Calculated LOAD value	%	
Engine coolant temperature	°C	°F
Intake manifold absolute pressure	kPa	
Engine speed	rpm	
Vehicle speed	km/h	mph
Intake air temperature	°C	°F
Air flow rate from mass air flow sensor	g/s	
OBD requirement according to vehicle design	—	
Time Since Engine Start	s	
Distance travelled while MIL is activated	km	miles
Fuel rail pressure	kPa	
EGR valve control signal	%	
Number of warm-ups since DTCs cleared	—	
Distance travelled since DTCs cleared	km	miles
Barometric Pressure	kPa	
PCM power supply voltage	V	
Ambient air temperature	°C	°F
Accelerator pedal position (sensor No.1)	%	
Accelerator pedal position (sensor No.2)	%	

Sending Freeze Frame Data

- The Freeze Frame Data monitor items are shown below.

Freeze Frame Data monitor table

—: Not applicable

Full names	Unit	
DTC that caused required Freeze Frame Data storage	—	
Calculated LOAD value	%	
Engine coolant temperature	°C	°F
Intake manifold absolute pressure	kPa	
Engine speed	rpm	
Vehicle speed	km/h	mph
Intake air temperature	°C	°F
Air flow rate from mass air flow sensor	g/s	
Fuel rail pressure	kPa	
Barometric Pressure	kPa	
Accelerator pedal position (sensor No.1)	%	
Accelerator pedal position (sensor No.2)	%	

ON-BOARD DIAGNOSTIC [WL-C, WE-C]

Sending Emission-related Malfunction Code

The DTCs are shown below.

: Applicable
—: Not applicable

DTC	Condition	MIL	DC	Self-test type*1	Memory function
P0016	CKP-CMP correlation	ON	1	C/R	
P0045	Variable boost control (VBC) solenoid valve control circuit open	ON	1	C/O/R	
P0046	Variable boost control (VBC) solenoid valve circuit range/performance malfunction	ON	1	C/O/R	
P0047	Variable boost control (VBC) solenoid valve control circuit low input	ON	1	C/O/R	
P0048	Variable boost control (VBC) solenoid valve control circuit high input	ON	1	C/O/R	
P0088	Fuel pressure system too high	ON	1	C	
P0093	Fuel system leak detection	ON	1	C	
P0097	Intake air temperature (IAT) sensor No.2 circuit low input	ON	1	C/O/R	
P0098	Intake air temperature (IAT) sensor No.2 circuit high input	ON	1	C/O/R	
P0101	Mass airflow (MAF) sensor range/performance problem	ON	2	—	
P0102	Mass airflow (MAF) sensor circuit low input	ON	1	C/R	
P0103	Mass airflow (MAF) sensor circuit high input	ON	1	C/R	
P0106	Boost sensor range/performance malfunction	ON	1	C	
P0107	Boost sensor circuit low input	ON	1	C/O/R	
P0108	Boost sensor circuit high input	ON	1	C/O/R	
P0111	Intake air temperature (IAT) sensor No.1 range/performance	ON	2	C	
P0112	Intake air temperature (IAT) sensor No.1 circuit low input	ON	1	C/O/R	
P0113	Intake air temperature (IAT) sensor No.1 circuit high input	ON	1	C/O/R	
P0116	Engine coolant temperature (ECT) sensor range/performance problem	ON	2	—	
P0117	Engine coolant temperature (ECT) sensor circuit low input	ON	1	C/O/R	
P0118	Engine coolant temperature (ECT) sensor circuit high input	ON	1	C/O/R	
P0122	Accelerator pedal position (APP) sensor No.1 circuit low input	ON	1	C/O/R	
P0123	Accelerator pedal position (APP) sensor No.1 circuit high input	ON	1	C/O/R	
P0182	Fuel temperature sensor circuit low input	ON	1	C/O/R	
P0183	Fuel temperature sensor circuit high input	ON	1	C/O/R	
P0191	Fuel pressure sensor range/performance malfunction	ON	1	O	
P0192	Fuel pressure sensor circuit low input	ON	1	C/O/R	
P0193	Fuel pressure sensor circuit high input	ON	1	C/O/R	
P0201	Fuel injector No.1 circuit open/short	ON	1	C/R	
P0202	Fuel injector No.2 circuit open/short	ON	1	C/R	
P0203	Fuel injector No.3 circuit open/short	ON	1	C/R	
P0204	Fuel injector No.4 circuit open/short	ON	1	C/R	
P0222	Accelerator pedal position (APP) sensor No.2 circuit low input	ON	1	C/O/R	
P0223	Accelerator pedal position (APP) sensor No.2 circuit high input	ON	1	C/O/R	
P0227	Idle switch circuit low input	ON	1	C	
P0228	Idle switch circuit high input	ON	1	C	
P0300	Random misfire detected	ON	2	C/R	
P0301	Cylinder No.1 misfire detection	ON	2	C/R	
P0302	Cylinder No.2 misfire detection	ON	2	C/R	
P0303	Cylinder No.3 misfire detection	ON	2	C/R	
P0304	Cylinder No.4 misfire detection	ON	2	C/R	
P0335	Crankshaft position (CKP) sensor circuit malfunction	ON	1	C/R	
P0340	Camshaft position (CMP) sensor circuit malfunction	ON	1	C/R	
P0380	Glow plug relay circuit malfunction	ON	1	C/O	
P0401	EGR flow insufficient detected	ON	2	C	
P0402	EGR flow excessive detected	ON	2	C	

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ON-BOARD DIAGNOSTIC [WL-C, WE-C]

DTC	Condition	MIL	DC	Self-test type*1	Memory function
P0403	EGR solenoid valve circuit malfunction	ON	1	C/O/R	
P0404	EGR valve stuck	ON	2	C/R	
P0405	EGR valve position sensor circuit low input	ON	1	C/O/R	
P0406	EGR valve position sensor circuit high input	ON	1	C/O/R	
P0489	EGR solenoid valve control circuit low input	ON	1/2	C/O	
P0490	EGR solenoid valve control circuit high input	ON	1/2	C/O	
P0500	VSS circuit malfunction	ON	2	C	
P0562	Battery voltage low input	ON	1	C/O/R	
P0563	Battery voltage high input	ON	1	C/O/R	
P0602	PCM programming error	ON	1	O	
P0606	PCM malfunction	ON	1	C/O/R	
P0607	PCM performance problem	ON	1/2	C/O/R	
P0610	PCM vehicle options error	ON	1	O	
P0627	Fuel metering valve control circuit open	ON	1	C/R	
P0628	Fuel metering valve control circuit low input	ON	1	C/R	
P0629	Fuel metering valve control circuit high input	ON	1	C/R	
P0642	Fuel pressure sensor and boost sensor 5 V circuit low input	ON	1	C/O/R	
P0643	Fuel pressure sensor and boost sensor 5 V circuit high input	ON	1	C/O/R	
P0652	CMP sensor and APP sensor 5 V circuit low input	ON	1	C/O/R	
P0653	CMP sensor and APP sensor 5 V circuit high input	ON	1	C/O/R	
P0660	Intake shutter solenoid valve (half) control circuit open	ON	1	C/O	
P0661	Intake shutter solenoid valve (half) control circuit low input	ON	1	C/O	
P0662	Intake shutter solenoid valve (half) control circuit high input	ON	1	C/O	
P0663	Intake shutter solenoid valve (full) control circuit open	ON	1	C/O	
P0664	Intake shutter solenoid valve (full) control circuit low input	ON	1	C/O	
P0665	Intake shutter solenoid valve (full) control circuit high input	ON	1	C/O	
P0685	Main relay control circuit open	—	—	—	
P0698	EGR valve position sensor 5 V circuit low input	ON	1	C/O/R	
P0699	EGR valve position sensor 5 V circuit high input	ON	1	C/O/R	
P1196	Key off voltage high input	—	—	—	
P1259	IMMOBILIZER to PCM signal error	—	—	C/O	—
P1260	Theft detected Vehicle immobilizer	—	—	C/O	—
P1391	Glow plug control circuit low input	ON	1	C/O	
P1392	Glow plug control circuit high input	ON	1	C/O	
P1528	Exhaust shutter solenoid valve circuit problem	—	1	C/O	—
P1602	Immobilizer/PCM communication error	—	—	C/O	—
P1603	ID Number Unregistered	—	—	C/O	—
P1604	Code word unregistered	—	—	C/O	—
P1621	Immobilizer code word does not match	—	—	C/O	—
P1622	Immobilizer ID does not match	—	—	C/O	—
P1623	Immobilizer code word/ID number write failure	—	—	C/O	—
P1624	Anti-theft system	—	—	C/O	—
P1675	Injection quantity adjustment value writing error	ON	1	O	
P1676	Injection quantity adjustment value checksum error	ON	1	O	
P2008	Variable swirl control (VSC) solenoid valve circuit open	ON	1	C/O	
P2009	Variable swirl control (VSC) solenoid valve circuit low input	ON	1	C/O	
P2010	Variable swirl control (VSC) solenoid valve circuit high input	ON	1	C/O	
P2135	Accelerator pedal position (APP) sensor No.1/No.2 voltage correlation problem	ON	1	C/O/R	
P2136	Accelerator pedal position (APP) sensor No.1/idle switch voltage correlation problem	ON	1	C/O/R	
P2143	EGR control solenoid valve control circuit open	ON	1/2	C/O	
P2144	EGR control solenoid valve control circuit low input	ON	1/2	C/O/R	
P2145	EGR control solenoid valve control circuit high input	ON	1/2	C/O/R	

ON-BOARD DIAGNOSTIC [WL-C, WE-C]

DTC	Condition	MIL	DC	Self-test type*1	Memory function
P2146	Fuel injector No.1 and No.4 power supply circuit malfunction	ON	1	C/R	
P2147	Fuel injector No.1 and No.4 power supply circuit low input	ON	1	C/R	
P2148	Fuel injector No.1 and No.4 power supply circuit high input	ON	1	C/R	
P2149	Fuel injector No.2 and No.3 power supply circuit malfunction	ON	1	C/R	
P2150	Fuel injector No.2 and No.3 power supply circuit low input	ON	1	C/R	
P2151	Fuel injector No.2 and No.3 power supply circuit high input	ON	1	C/R	
P2227	BARO sensor circuit range/performance malfunction	ON	1	C	
P2228	BARO sensor circuit low input	ON	1	C/O/R	
P2229	BARO sensor circuit high input	ON	1	C/O/R	
P2263	Air charging system performance malfunction	—	1	C	
P2530	PCM (engine switch circuit) malfunction	ON	1	O	

*1 : C; CMDTC self test, O; KOEO self test, R; KOER self test

Sending Continuous Monitoring System Test Results

These appear when a problem is detected in a monitored system.

1-drive cycle type

If any problems are detected in the first drive cycle, confirmed DTCs will be stored in the PCM memory.

2-drive cycle type

The code for a failed system is stored in the PCM memory in the first drive cycle. If the problem is not found in the second drive cycle, the PCM determines that the system returned to normal or the problem was mistakenly detected, and deletes the pending code. If the problem is found in the second drive cycle too, the PCM determines that the system has failed, and stores the DTCs.

After pending codes are stored, if the PCM determines that the system is normal in any future drive cycle, the PCM deletes the pending codes.

DTC DETECTION LOGIC AND CONDITIONS[WL-C, WE-C]

id0102c4100300

P0016 CKP-CMP correlation

The PCM monitors the input pulses from the CKP and CMP sensors. If the pick-up timing input pulses do not match each other, the PCM determines that the camshaft position does not coincide with the crankshaft position.

P0045 Variable boost control (VBC) solenoid valve control circuit open

The PCM monitors the VBC solenoid valve control signal. If the VBC solenoid valve control voltage is **less than 5 V for 2 s** when the VBC solenoid valve is off, the PCM determines that there is a VBC solenoid valve control circuit malfunction (open circuit).

P0046 Variable boost control (VBC) solenoid valve circuit range/performance malfunction

The PCM monitors the driver IC temperature of the VBC solenoid valve. If the temperature of the VBC solenoid valve driver IC in the PCM exceeds **150 °C{320 °F}** while the VBC solenoid valve is operating, the PCM determines that there is a VBC solenoid valve malfunction.

P0047 Variable boost control (VBC) solenoid valve control circuit low input

The PCM monitors the VBC solenoid valve control signal. If the PCM turns VBC solenoid valve off but the voltage at PCM terminal 178 still remains low, the PCM determines that the VBC solenoid valve circuit has a malfunction (circuit short to ground).

P0048 Variable boost control (VBC) solenoid valve control circuit high input

The PCM monitors the VBC solenoid valve control signal. If the PCM turns VBC solenoid valve on but the current at PCM terminal 178 still remains high, the PCM determines that the VBC solenoid valve circuit has a malfunction (circuit short to power supply).

P0088 Fuel pressure system too high

The PCM monitors the fuel pressure in the common rail from the fuel pressure sensor while the engine is running. If the fuel pressure is more than the preprogrammed criteria, the PCM determines that the fuel pressure is too high.

P0093 Fuel system leak detection

The PCM monitors the fuel pressure in the common rail from the fuel pressure sensor while the engine is running. If the fuel pressure is lower than the preprogrammed criteria, the PCM determines that there is leakage in the fuel system.

ON-BOARD DIAGNOSTIC [WL-C, WE-C]

P0097 Intake air temperature (IAT) sensor No.2 circuit low input

The PCM monitors the input signal from intake air temperature sensor No.2. If the voltage from intake air temperature sensor No.2 is **less than 0.1 V for 2 s**, the PCM determines that there is a malfunction in the intake air temperature sensor No.2 circuit.

P0098 Intake air temperature (IAT) sensor No.2 circuit high input

The PCM monitors the input signal from intake air temperature sensor No.2. If the voltage from intake air temperature sensor No.2 is **more than 4.6 V for 1 s**, the PCM determines that there is a malfunction in the intake air temperature sensor No.2 circuit.

P0101 Mass airflow (MAF) sensor range/performance malfunction

The PCM monitors the ratio between calculated air mass flow and actual air mass flow under the programmed condition.

If the ratio is more than or less than the programmed criteria the PCM determine that there is a malfunction in the MAF sensor.

P0102 Mass airflow (MAF) sensor circuit low input

The PCM calculates the intake air flow amount based on the airflow sensor. If the calculated intake air flow amount is **below 0.44 g/s {0.058 lb/min} for 2.0 s** while the engine is running, the PCM determines that the airflow sensor signal input is low.

P0103 Mass airflow (MAF) sensor circuit high input

The PCM calculates the intake air flow amount based on the airflow sensor. If the calculated intake air flow amount is **above 170 g/s {22 lb/min} for 1.5 s** while driving other than under a fully open throttle condition, and the fuel injection quantity based on the engine speed is below the specification, the PCM determines that the airflow sensor signal input is high.

P0106 Boost sensor range/performance malfunction

The PCM monitors and compares the intake manifold internal pressure via the boost sensor with the atmospheric pressure via the barometric pressure sensor (built into PCM) while the engine is not running. If the difference of these pressure is **above 11 kPa {0.11 bar, 1.6 psi}**, the PCM determines that there is a boost sensor characteristic malfunction.

P0107 Boost sensor circuit low input

The PCM monitors the input signal from the boost sensor. If the voltage from the boost sensor is **less than 0.24 V for 2.0 s**, the PCM determines that there is a malfunction in the boost sensor circuit.

P0108 Boost sensor circuit high input

The PCM monitors the input signal from the boost sensor. If the voltage from the boost sensor is **more than 4.9 V for 1.0 s**, the PCM determines that there is a malfunction in the boost sensor circuit.

P0111 Intake air temperature (IAT) sensor No.1 range/performance malfunction

The PCM monitors the input signal from the IAT sensor No.1. If the difference between the maximum and minimum value of the IAT sensor No.1 is less than 1 °C {1.8 °F}, the PCM determines that there is a malfunction in the IAT sensor No.1.

P0112 Intake air temperature (IAT) sensor No.1 circuit low input

The PCM monitors the input signal from IAT sensor No.1. If the voltage from IAT sensor No.1 is **less than 0.14 V for 2.0 s**, the PCM determines that there is a malfunction in the IAT sensor No.1 circuit.

P0113 Intake air temperature (IAT) sensor No.1 circuit high input

The PCM monitors the input signal from IAT sensor No.1. If the voltage from IAT sensor No.1 is **more than 4.90 V for 1.0 s**, the PCM determines that there is a malfunction in the IAT sensor No.1 circuit.

P0116 Engine coolant temperature (ECT) sensor range/performance malfunction

Minimum engine coolant temperature rise is not achieved within the specified time limit from when the engine was started.

P0117 Engine coolant temperature (ECT) sensor circuit low input

The PCM monitors the input signal from the ECT sensor. If the voltage from the ECT sensor is **less than 0.20 V for 2.0 s**, the PCM determines that there is a malfunction in the ECT sensor circuit.

P0118 Engine coolant temperature (ECT) sensor circuit high input

The PCM monitors the input signal from the ECT sensor. If the voltage from the ECT sensor is **more than 4.91 V for 1.0 s**, the PCM determines that there is a malfunction in the ECT sensor circuit.

P0122 Accelerator pedal position (APP) sensor No.1 circuit low input

The PCM monitors the input signal from APP sensor No.1. If the voltage from APP sensor No.1 is **less than 0.29 V for 0.5 s** when the APP sensor No.1 power supply voltage is normal, the PCM determines that there is a malfunction in the APP sensor No.1 circuit.

P0123 Accelerator pedal position (APP) sensor No.1 circuit high input

The PCM monitors the input signal from APP sensor No.1. If the voltage from APP sensor No.1 is **more than 4.80 V for 0.5 s** when the APP sensor No.1 power supply voltage is normal, the PCM determines that there is a malfunction in the APP sensor No.1 circuit.

P0182 Fuel temperature sensor circuit low input

The PCM monitors the input signal from the fuel temperature sensor. If the voltage from the fuel temperature sensor is **less than 0.2 V**, the PCM determines that there is a malfunction in the fuel temperature sensor circuit.

P0183 Fuel temperature sensor circuit high input

The PCM monitors the input signal from the fuel temperature sensor. If the voltage from the fuel temperature sensor is **more than 4.9 V**, the PCM determines that there is a malfunction in the fuel temperature sensor circuit.

P0191 Fuel pressure sensor range/performance malfunction

The PCM monitors the fuel pressure in the common rail and input signal from the fuel pressure sensor while the engine is running. If all of the following conditions is met, the PCM determines that there is malfunction in the fuel pressure sensor range/performance malfunction.

- The PCM calculates the difference between the actual fuel pressure and the target fuel pressure. If the pressure difference is **more than 5 MPa {51 kgf/cm², 726 psi}**.
- The PCM monitors the input signal from the fuel pressure sensor. If the difference between the maximum and minimum voltage of the fuel pressure sensor is **less than 0.01 V**.

P0192 Fuel pressure sensor circuit low input

The PCM monitors the input voltage from the fuel pressure sensor while the engine is running. If the input voltage from the fuel pressure sensor is **less than 0.2 V**, the PCM determines that there is a malfunction in the fuel pressure sensor circuit.

P0193 Fuel pressure sensor circuit high input

The PCM monitors the input voltage from the fuel pressure sensor while the engine is running. If the input voltage from the fuel pressure sensor is **more than 4.8 V**, the PCM determines that there is a malfunction in the fuel pressure sensor circuit.

P0201 Fuel injector No.1 circuit open/short

The PCM monitors each cylinder fuel injector control signal current while the engine is running. If the PCM detects a control signal current of more or less than the preprogrammed criteria, the PCM determines that the fuel injector at No. 1 cylinder has an open or short circuit.

P0202 Fuel injector No.2 circuit open/short

The PCM monitors each cylinder fuel injector control signal current while the engine is running. If the PCM detects a control signal current of more or less than the preprogrammed criteria, the PCM determine that the fuel injector at No.2 cylinder has an open or short circuit.

P0203 Fuel injector No.3 circuit open/short

The PCM monitors each cylinder fuel injector control signal current while the engine is running. If the PCM detects a control signal current of more or less than the preprogrammed criteria, the PCM determines that the fuel injector at No.3 cylinder has an open or short circuit.

P0204 Fuel injector No.4 circuit open/short

The PCM monitors each cylinder fuel injector control signal current while the engine is running. If the PCM detects a control signal current of more or less than the preprogrammed criteria, the PCM determines that the fuel injector at No.4 cylinder has an open or short circuit.

P0222 Accelerator pedal position (APP) sensor No.2 circuit low input

The PCM monitors the input signal from APP sensor No.2. If the voltage from APP sensor No.2 is **less than 0.29 V for 0.5 s** when the APP sensor No.2 power supply voltage is normal, the PCM determines that there is a malfunction in the APP sensor No.2 circuit.

P0223 Accelerator pedal position (APP) sensor No.2 circuit high input

The PCM monitors the input signal from APP sensor No.2. If the voltage from APP sensor No.2 is **more than 4.80 V for 0.5 s** when the APP sensor no.2 power supply voltage is normal, the PCM determines that there is a malfunction in the APP sensor No.2 circuit.

P0227 Idle switch circuit low input

The PCM monitors the input voltage from APP sensor No.1 when the idle switch is off. If the input voltage is **less than 0.86 V for 1.5 s**, the PCM determines that the idle switch circuit has a malfunction.

P0228 Idle switch circuit high input

The PCM monitors the input voltage from APP sensor No.1 when the idle switch is on. If the input voltage is **more than 1.35 V for 1.5 s**, the PCM determines that the idle switch circuit has a malfunction.

P0300 Random misfire detected

The PCM monitors the CKP sensor input signal interval time. The PCM calculates the deviation of the interval time for each cylinder. If the deviation of the interval time exceeds the preprogrammed criteria, the PCM detects a misfire in the corresponding cylinder. While the engine is running, the PCM counts the number of misfires and calculates the misfire ratio for each crankshaft revolution. If the ratio exceeds the preprogrammed criteria, the PCM determines that a misfire, which can affect emission performance, has occurred.

P0301, P0302, P0303, P0304 Cylinder No.1 No.2 No.3 No.4 misfire detection

The PCM monitors the CKP sensor input signal interval time. The PCM calculates the deviation of the interval time for each cylinder. If the deviation of the interval time exceeds the preprogrammed criteria, the PCM detects a misfire in the corresponding cylinder. While the engine is running, the PCM counts the number of misfires and calculates the misfire ratio for each crankshaft revolution. If the ratio exceeds the preprogrammed criteria, the PCM determines that a misfire, which can affect emission performance, has occurred.

ON-BOARD DIAGNOSTIC [WL-C, WE-C]

P0335 Crankshaft position (CKP) sensor circuit malfunction

The PCM monitors the input signal from the CKP sensor while the engine is running. If the following conditions are detected, the PCM determines that the CKP circuit has a malfunction.

- The PCM receives no pulse from the CKP sensor
- The PCM receives an improper pulse number from the CKP sensor

P0340 Camshaft position (CMP) sensor circuit malfunction

The PCM monitors the input signal from the CMP sensor while the engine is running. If the following conditions are detected, the PCM determines that the CMP circuit has a malfunction.

- The PCM receives no pulse from the CMP sensor
- The PCM receives an improper pulse number from the CMP sensor

P0380 Glow plug relay circuit malfunction

The PCM monitors the glow plug relay control signal. If the following conditions are detected, the PCM determines that the glow plug relay circuit has a malfunction.

- The PCM turns the glow plug relay off but the voltage remains low
- The PCM turns the glow plug relay on but the voltage remains high
- The PCM internal driver IC temperature is **more than 150 °C {320 °F}** when the glow plug relay is on

P0401 EGR flow insufficient detected

The PCM monitors the difference between the target air amount and intake air amount while the EGR system is operating at EGR feedback control. If the difference between the target air amount and the intake air amount is less than the threshold, the PCM determines that there is a malfunction in the EGR system.

P0402 EGR flow excessive detected

The PCM monitors the difference between the target air amount and intake air amount while the EGR system is operating at EGR feedback control. If the difference between the target air amount and the intake air amount is more than the threshold, the PCM determines that there is a malfunction in the EGR system.

P0403 EGR solenoid valve circuit malfunction

The PCM monitors the EGR solenoid valve control signal. If the following conditions are detected, the PCM determines that the EGR solenoid valve circuit has a malfunction (open circuit).

- The PCM turns the EGR solenoid valve off but the voltage remains low
- The PCM internal driver IC temperature is **more than 150 °C {320 °F}** when the EGR solenoid valve is on

P0404 EGR control circuit range/performance

The PCM monitors the difference between target EGR valve lift and actual EGR valve lift under the programmed condition.

If the difference is more or less than the programmed criteria the PCM determines that there is a malfunction in the EGR control circuit.

P0405 EGR valve position sensor circuit low input

The PCM monitors the input voltage from the EGR valve position sensor while the engine is running. If the input voltage is **less than 0.1 V**, the PCM determines that the EGR valve position sensor circuit has a malfunction.

P0406 EGR valve position sensor circuit high input

The PCM monitors the input voltage from the EGR valve position sensor while the engine is running. If the input voltage is **more than 4.9 V**, the PCM determines that the EGR valve position sensor circuit has a malfunction.

P0489 EGR solenoid valve control circuit low input

The PCM monitors the EGR solenoid valve control signal. If the PCM turns the EGR solenoid valve off but the voltage remains low, the PCM determines that EGR solenoid valve circuit has a malfunction (circuit short to ground).

P0490 EGR solenoid valve control circuit high input

The PCM monitors the EGR solenoid valve control signal. If the PCM turns the EGR solenoid valve on but the voltage remains high, the PCM determines that EGR solenoid valve circuit has a malfunction (circuit short to power supply).

P0500 VSS circuit malfunction

The PCM monitors the input signal from the instrument cluster during set conditions. If the input signal is **less than 0 km/h {0 mph} for 25 s**, the PCM determines that there is a VSS circuit malfunction.

P0562 Battery voltage low input

The PCM monitors the battery voltage. If the battery voltage is **less than 8 V**, the PCM determines that there is a malfunction in the battery and the battery signal system.

P0563 Battery voltage high input

The PCM monitors the battery voltage. If the battery voltage is **more than 16 V**, the PCM determines that there is a malfunction in the battery charging system.

P0602 PCM programming error

No configuration data in the PCM

P0606 PCM malfunction

PCM internal malfunction.

P0607 PCM performance problem

PCM internal malfunction.

P0610 PCM vehicle options error

PCM data configuration error

P0627 Fuel metering valve control circuit open

The PCM monitors the fuel metering valve driver IC temperature in the PCM and the control voltage while the engine is running. If either of the following conditions is met, the PCM determines that there is a malfunction in the fuel metering valve range/performance.

- The PCM monitors the fuel metering valve driver IC temperature in the PCM. If the fuel metering valve driver IC temperature is **more than 150 C {320 F}** during the fuel metering valve operation, the PCM determines that the fuel metering valve has a malfunction
- The PCM monitors the fuel metering valve control voltage at the PCM terminals. If fuel metering valve control voltage is **less than 5 V** when fuel metering valve is not operating, the PCM determines that the fuel metering valve has a malfunction (open circuit)

P0628 Fuel metering valve control circuit low input

The PCM monitors the fuel metering valve circuit voltage while the engine is running. If the PCM detects that the circuit voltage is low when the fuel metering valve is off, the PCM determines that there is a malfunction in the fuel metering valve circuit (circuit short to ground).

P0629 Fuel metering valve control circuit high input

The PCM monitors the fuel metering valve circuit voltage while the engine is running. If the PCM detects that the circuit voltage is high when the fuel metering valve is on, the PCM determines that there is a malfunction in the fuel metering valve circuit (circuit short to power supply).

P0642 Fuel pressure sensor and boost sensor 5 V circuit low input

The PCM monitors the fuel pressure sensor and the boost sensor 5 V circuit. If the fuel pressure sensor and boost sensor power voltage is **lower than 4.9 V**, the PCM determines that there is a fuel pressure sensor and boost sensor 5 V circuit malfunction.

P0643 Fuel pressure sensor and boost sensor 5 V circuit high input

The PCM monitors the fuel pressure sensor and boost sensor 5 V circuit. If the fuel pressure sensor and boost sensor power voltage is **higher than 4.9 V**, the PCM determines that there is a fuel pressure sensor and boost sensor 5 V circuit malfunction.

P0652 CMP sensor and APP sensor 5 V circuit low input

The PCM monitors the CMP sensor and APP sensor 5 V circuit. If the CMP sensor and APP sensor power voltage is **lower than 4.9 V**, the PCM determines that there is a CMP sensor and APP sensor 5 V circuit malfunction.

P0653 CMP sensor and APP sensor 5 V circuit high input

The PCM monitors the CMP sensor and APP sensor 5 V circuit. If the CMP sensor and APP sensor power voltage is **higher than 5.1 V**, the PCM determines that there is a CMP sensor and APP sensor 5 V circuit malfunction.

P0660 Intake shutter solenoid valve (half) control circuit open

The PCM monitors the intake shutter solenoid valve (half) control signal. If the following conditions are detected, the PCM determines that the intake shutter solenoid valve (half) control circuit has a malfunction.

- The PCM turns the intake shutter solenoid valve (half) off, but the intake shutter solenoid valve (half) control voltage is **less than 5 V for 2 s** (open circuit)
- The PCM internal driver IC temperature is **more than 150 C {320 F}** when the intake shutter solenoid valve (half) is on

P0661 Intake shutter solenoid valve (half) control circuit low input

The PCM monitors the intake shutter solenoid valve (half) control signal at PCM terminal 152. If the PCM turns the intake shutter solenoid valve (half) off but the voltage at PCM terminal 152 remains low, the PCM determines that intake shutter solenoid valve (half) circuit has a malfunction (circuit short to ground).

P0662 Intake shutter solenoid valve (half) control circuit high input

The PCM monitors the intake shutter solenoid valve (half) control signal at PCM terminal 152. If the PCM turns the intake shutter solenoid valve (half) on but the voltage at PCM terminal 152 remains high, the PCM determines that the intake shutter solenoid valve (half) circuit has a malfunction (circuit short to power supply).

P0663 Intake shutter solenoid valve (full) control circuit open

The PCM monitors the intake shutter solenoid valve (full) control signal. If the following conditions are detected, the PCM determines that the intake shutter solenoid valve (full) control circuit has a malfunction.

- The PCM turns the intake shutter solenoid valve (full) off, but the intake shutter solenoid valve (full) control voltage is **less than 5 V for 2 s** (open circuit)
- The PCM internal driver IC temperature is **more than 150 C {320 F}** when the intake shutter solenoid valve (full) is on

P0664 Intake shutter solenoid valve (full) control circuit low input

The PCM monitors the intake shutter solenoid valve (full) control signal at PCM terminal 155. If the PCM turns the intake shutter solenoid valve (full) off, but the voltage at PCM terminal 155 remains low, the PCM determines that intake shutter solenoid valve (full) circuit has a malfunction (circuit short to ground).

P0665 Intake shutter solenoid valve (full) control circuit high input

The PCM monitors the intake shutter solenoid valve (full) control signal at PCM terminal 155. If the PCM turns the intake shutter solenoid valve (full) on, but the voltage at PCM terminal 155 remains high, the PCM determines that the intake shutter solenoid valve (full) circuit has a malfunction (circuit short to power supply).

ON-BOARD DIAGNOSTIC [WL-C, WE-C]

P0685 Main relay control circuit open

The PCM monitors the power supply to the PCM. If the power supply is shut down several times when the engine switch is turned to the ON position, the PCM determines that there is a power circuit malfunction (open circuit).

P0698 EGR valve position sensor 5 V circuit low input

The PCM monitors the EGR position sensor 5 V circuit. If the EGR position sensor power voltage is **lower than 4.9 V**, the PCM determines that there is a EGR position sensor 5 V circuit malfunction.

P0699 EGR valve position sensor 5 V circuit high input

The PCM monitors the EGR position sensor 5 V circuit. If the EGR position sensor power voltage is **higher than 5.1 V**, the PCM determines that there is a EGR position sensor 5 V circuit malfunction.

P1196 Key off voltage high input

The PCM monitors whether the main relay opens **after 2 s** when the engine switch is turned to the off position. If the main relay does not open **after 2 s**, the PCM determines that there is malfunction in the main relay circuit.

P1259 IMMOBILIZER to PCM signal error

The PCM does not complete target ID transfer.

P1260 Theft detected Vehicle immobilizer

Immobilizer system malfunction

P1391 Glow plug control circuit low input

The PCM monitors the output signal to the glow plug relay when the glow plug relay is on. If the glow plug relay voltage is **less than 2 V for 1 s** when the glow plug relay is operating, The PCM determines that there is a malfunction in the glow plug relay circuit.

P1392 Glow plug control circuit high input

The PCM monitors the output signal to the glow plug relay when the glow plug relay is off. If the glow plug relay voltage is **more than 2 V for 1 s** when the glow plug relay is off, The PCM determines that there is a malfunction in the glow plug relay control circuit.

P1528 Exhaust shutter solenoid valve circuit problem

The PCM monitors the input voltages from the exhaust shutter solenoid valve. If the voltage remains low or high, the PCM determines that the exhaust shutter solenoid valve circuit has malfunction.

P1602 Immobilizer/PCM communication error

PCM internal malfunction

P1603 ID Number Unregistered

PCM internal malfunction

P1604 Code word unregistered

PCM internal malfunction

P1621 Immobilizer code word does not match

Cord word stored in immobilizer system and PCM does not to match.
ID number stored in immobilizer system and PCM does not to match.

P1622 Immobilizer ID does not match

Cord word stored in immobilizer system and PCM does not to match.
ID number stored in immobilizer system and PCM does not to match.

P1623 Immobilizer code word/ID number write failure

Cord word stored in immobilizer system and PCM does not to match.
ID number stored in immobilizer system and PCM does not to match.

P1624 Anti-theft system

Cord word stored in immobilizer system and PCM does not to match.
ID number stored in immobilizer system and PCM does not to match.

P1675 Injection quantity adjustment value writing error

The fuel injector compensation data configuration has not been completed.

P1676 Injection quantity adjustment value checksum error

The fuel injector compensation data check sum error.

P2008 Variable swirl control (VSC) solenoid valve control circuit open

The PCM monitors the VSC solenoid valve control signal. If the following conditions are detected, the PCM determines that the VSC solenoid valve control circuit has a malfunction.

- The PCM turns the VSC solenoid valve off, but the VSC solenoid valve control voltage is **less than 5 V for 2 s** (open circuit)
- The PCM internal driver IC temperature is **more than 150 C {320 F}** when the VSC solenoid valve is on

P2009 Variable swirl control (VSC) solenoid valve circuit low input

The PCM monitors the output signal to the VSC solenoid valve. If the voltage of the VSC solenoid valve is low even if the VSC solenoid valve is off, the PCM determines that there is a malfunction in the VSC solenoid valve control circuit.

P2010 Variable swirl control (VSC) solenoid valve circuit high input

The PCM monitors the output signal to the VSC solenoid valve. If the voltage of the VSC solenoid valve is high even if the VSC solenoid valve is on, the PCM determines that there is a malfunction in the VSC solenoid valve control system.

P2135 Accelerator pedal position (APP) sensor No.1/No.2 voltage correlation problem

The PCM monitors the input signals from accelerator pedal position sensor No.1 and accelerator pedal position sensor No.2. If the difference between accelerator pedal position sensor No.1 and accelerator pedal position sensor No.2 is **more than 0.9 V for 0.5 s**, the PCM determines that there is a malfunction in the accelerator pedal position sensor characteristic.

P2136 Accelerator pedal position (APP) sensor No.1/idle switch voltage correlation problem

The PCM compares the input voltage from APP sensor No.1 with the input voltage from the idle switch when the accelerator pedal is released (idle switch is on). If the difference is **more than 1.35 V for 0.5 s**, the PCM determines that there is an APP sensor No.1/idle switch angle correlation problem.

P2143 EGR control solenoid valve control circuit open

The PCM monitors the EGR control solenoid valve control signal. If the following conditions are detected, the PCM determines that the EGR control solenoid valve circuit has a malfunction.

- The PCM turns the EGR control solenoid valve off but the voltage remains low (open circuit).
- The PCM internal driver IC temperature is more than **150 C {320 F}** when the EGR control solenoid valve is on.

P2144 EGR control solenoid valve control circuit low input

The PCM monitors the EGR control solenoid valve control signal. If the PCM turns the EGR control solenoid valve off but the voltage remains low, the PCM determines that the EGR control solenoid valve circuit has a malfunction (circuit short to ground).

P2145 EGR control solenoid valve control circuit high input

The PCM monitors the EGR control solenoid valve control signal. If the PCM turns the EGR control solenoid valve on but the voltage still remains high, the PCM determines that the EGR control solenoid valve circuit has a malfunction (circuit short to power supply).

P2146 Fuel injector No.1 and No.4 power supply circuit malfunction

The PCM monitors the fuel injector No.1 and No.4 power supply voltage when the engine is running. If the following conditions are detected 5 times, the PCM determines that the fuel injector No.1 and No.4 power supply circuit has a malfunction.

- Fuel injector No.1 and No.4 power supply voltage is **less than 43 V**
- Fuel injector No.1 and No.4 power supply voltage is **53 V or above**

P2147 Fuel injector No.1 and No.4 power supply circuit low input

The PCM monitors the fuel injector No.1 and No.4 power supply current when the engine is running. If the power supply current is **25 A or above** 3 times while other fuel injector power supply current is **25 A or above**, the PCM determines that the fuel injector No.1 and No.4 power supply current is too high.

P2148 Fuel injector No.1 and No.4 power supply circuit high input

The PCM monitors the fuel injector No.1 and No.4 power supply current when the engine is running. If the power supply current is **50 A or above** 3 times, the PCM determines that the fuel injector No.1 and No.4 power supply current is too high.

P2149 Fuel injector No.2 and No.3 power supply circuit malfunction

The PCM monitors the fuel injector No.2 and No.3 power supply voltage when the engine is running. If the following conditions are detected 5 times, the PCM determines that the fuel injector No.2 and No.3 power supply circuit has a malfunction.

- Fuel injector No.2 and No.3 power supply voltage is **less than 43 V**
- Fuel injector No.2 and No.3 power supply voltage is **53 V or above**

P2150 Fuel injector No.2 and No.3 power supply circuit low input

The PCM monitors the fuel injector No.2 and No.3 power supply current when the engine is running. If the power supply current is **25 A or above** 3 times while other fuel injector power supply current is **25 A or above**, the PCM determines that the fuel injector No.2 and No.3 power supply current is too high.

P2151 Fuel injector No.2 and No.3 power supply circuit high input

The PCM monitors the fuel injector No.2 and No.3 power supply current when the engine is running. If the power supply current is **50 A or above** 3 times, the PCM determines that the fuel injector No.2 and No.3 power supply current is too high.

P2227 BARO sensor circuit range/performance malfunction

The PCM monitors the input signal from the barometric pressure sensor. If the difference between the barometric pressure input from the barometric pressure sensor and the boost pressure is **more than 11 kPa {0.11 bar, 1.6 psi}**, the PCM determines that there is a malfunction in the barometric pressure sensor.

P2228 BARO sensor circuit low input

The PCM monitors the input signal for the barometric pressure. If the voltage from the barometric pressure sensor is **less than 2.2 V for 1 s**, the PCM determines that there is a malfunction in the barometric pressure sensor signal system.

P2229 BARO sensor circuit high input

The PCM monitors the input signal for the barometric pressure. If the voltage from the barometric pressure sensor is **more than 4.8 V for 1 s**, the PCM determines that there is a malfunction in the barometric pressure sensor signal system.

ON-BOARD DIAGNOSTIC [WL-C, WE-C]

P2263 Air charging pressure performance malfunction

The PCM monitors the air charging pressure. If the air charging pressure is not within the specification for the PCM control target, the PCM determines that there is a malfunction in the air charging system.

P2530 PCM (engine switch circuit) malfunction

PCM internal malfunction.

PID/DATA MONITOR AND RECORD [WL-C, WE-C]

dcf01020000017

The PID/DATA monitor items are shown below.

PID/DATA monitor item table

Monitor item	Definition	Unit/Condition	PCM terminal
AAT	Ambient temperature	C	—
AC_REQ	A/C request signal	Off/On	239
ACCS	Air conditioning compressor cycling switch	Off/On	128
APP	Accelerator pedal position	%	209, 222
APP1	Accelerator pedal position sensor No.1	%	222
		V	
APP2	Accelerator pedal position sensor No.2	%	209
		V	
ARPMDES	Target engine speed	rpm	—
BARO	Barometric pressure	Pa	—
		V	
BOO	Brake switch	Off/On	238
CPP	Clutch pedal position switch	Off/On	228
DTCCNT	DTC count (includes those needing no action)	No unit	—
ECT	Engine coolant temperature	C	167
		V	
EGRV2	EGR solenoid valve (vent)	Off/On	177
FIP_FL	Supply pump flow control	A	179
		%	
FIP_SCV	Fuel metering valve	A	179
FLT	Fuel temperature	C	115
FRP	Fuel pressure sensor	Pa	140
GP_LMP	Glow indicator light	Off/On	233
GPC	Glow plug relay	Off/On	182
IASV	Intake shutter solenoid valve (half)	Off/On	152
IASV2	Intake shutter solenoid valve (full)	Off/On	155
IAT	Intake air temperature (IAT sensor No.1)	C	141
		V	
INGEAR	Load/no load condition	Off/On	240
IVS	Idle switch	Off Idle/Idle	161
LOAD	Engine load	%	—
MAF	Mass airflow amount	g/s	190
		V	
MAP	Boost sensor	Pa	189
		V	
MIL	Malfunction indicator lamp	Off/On	256
MIL_DIS	The distance travelled since the MIL was activated	km/h	—
RPM	Engine speed	rpm	193
SEGRP DSD	Desired EGR valve position	%	—
SELTESTDTC	Diagnostic trouble codes	No unit	—
VBCV	VBC solenoid valve	%	178
VPWR	Module supply voltage	V	201, 203, 205
VSS	Vehicle speed	km/h	184
WARM SW	Exhaust heating switch	Off/On	—

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ON-BOARD DIAGNOSTIC [WL-C, WE-C]

SIMULATION TEST [WL-C, WE-C]

dcf010200000t18

The simulation items are shown below.

Simulation item table

: Applicable
—: Not applicable

Item	Applicable component	Unit/condition	Test condition		PCM terminal
			KOEO	KOER	
ACCS	Air conditioning compressor cycling switch	Off/On		—	128
EGRV2	EGR solenoid valve (vent)	Off/On		—	177
GP_LMP	Glow indicator light	Off/On		—	233
GPC	Glow plug relay	Off/On		—	182
IASV	Intake shutter solenoid valve (half)	Off/On		—	152
IASV2	Intake shutter solenoid valve (full)	Off/On		—	155
INJ_1	Fuel injector No.1	Off	—		125, 174
INJ_2	Fuel injector No.2	Off	—		102, 149
INJ_3	Fuel injector No.3	Off	—		101, 173
INJ_4	Fuel injector No.4	Off	—		126, 150
SEGRP	EGR valve position sensor	%		—	116
VBCV	VBC solenoid valve	%		—	178
WARM_SOL	Exhaust shutter solenoid valve	Off/On		—	—

ON-BOARD DIAGNOSTIC [WL-C, WE-C]

KOEO/KOER SELF-TEST[WL-C, WE-C]

id0102c4100400

: Applicable
—: Not applicable

DTC	Condition	KOEO	KOER
P0016	CKP-CMP correlation	—	
P0045	Variable boost control (VBC) solenoid valve control circuit open		
P0046	Variable boost control (VBC) solenoid valve circuit range/performance malfunction		
P0047	Variable boost control (VBC) solenoid valve control circuit low input		
P0048	Variable boost control (VBC) solenoid valve control circuit high input		
P0088	Fuel pressure system too high	—	—
P0093	Fuel system leak detection	—	—
P0097	Intake air temperature (IAT) sensor No.2 circuit low input		
P0098	Intake air temperature (IAT) sensor No.2 circuit high input		
P0101	Mass airflow (MAF) sensor range/performance malfunction	—	—
P0102	Mass airflow (MAF) sensor circuit low input	—	
P0103	Mass airflow (MAF) sensor circuit high input	—	
P0106	Boost sensor range/performance malfunction	—	—
P0107	Boost sensor circuit low input		
P0108	Boost sensor circuit high input		
P0111	Intake air temperature (IAT) sensor No.1 range/performance malfunction	—	—
P0112	Intake air temperature (IAT) sensor No.1 circuit low input		
P0113	Intake air temperature (IAT) sensor No.1 circuit high input		
P0116	Engine coolant temperature (ECT) sensor range/performance malfunction	—	—
P0117	Engine coolant temperature (ECT) sensor circuit low input		
P0118	Engine coolant temperature (ECT) sensor circuit high input		
P0122	Accelerator pedal position (APP) sensor No.1 circuit low input		
P0123	Accelerator pedal position (APP) sensor No.1 circuit high input		
P0182	Fuel temperature sensor circuit low input		
P0183	Fuel temperature sensor circuit high input		
P0191	Fuel pressure sensor range/performance malfunction		—
P0192	Fuel pressure sensor circuit low input		
P0193	Fuel pressure sensor circuit high input		
P0201	Fuel injector No.1 circuit open/short	—	
P0202	Fuel injector No.2 circuit open/short	—	
P0203	Fuel injector No.3 circuit open/short	—	
P0204	Fuel injector No.4 circuit open/short	—	
P0222	Accelerator pedal position (APP) sensor No.2 circuit low input		
P0223	Accelerator pedal position (APP) sensor No.2 circuit high input		
P0227	Idle switch circuit low input	—	—
P0228	Idle switch circuit high input	—	—
P0300	Random misfire detected	—	
P0301	Cylinder No.1 misfire detection	—	
P0302	Cylinder No.2 misfire detection	—	
P0303	Cylinder No.3 misfire detection	—	
P0304	Cylinder No.4 misfire detection	—	
P0335	Crankshaft position (CKP) sensor circuit malfunction	—	
P0340	Camshaft position (CMP) sensor circuit malfunction	—	
P0380	Glow plug relay circuit malfunction		—
P0401	EGR flow insufficient detected	—	—

ON-BOARD DIAGNOSTIC [WL-C, WE-C]

DTC	Condition	KOEO	KOER
P0402	EGR flow excessive detected	—	—
P0403	EGR solenoid valve circuit malfunction		
P0404	EGR control circuit range/performance	—	
P0405	EGR valve position sensor circuit low input		
P0406	EGR valve position sensor circuit high input		
P0489	EGR solenoid valve control circuit low input		—
P0490	EGR solenoid valve control circuit high input		—
P0500	VSS circuit malfunction	—	—
P0562	Battery voltage low input		
P0563	Battery voltage high input		
P0602	PCM programming error		
P0606	PCM malfunction		
P0607	PCM performance problem		
P0610	PCM vehicle options error		
P0627	Fuel metering valve control circuit open	—	
P0628	Fuel metering valve control circuit low input	—	
P0629	Fuel metering valve control circuit high input	—	
P0642	Fuel pressure sensor and boost sensor 5 V circuit low input		
P0643	Fuel pressure sensor and boost sensor 5 V circuit high input		
P0652	CMP sensor and APP sensor 5 V circuit low input		
P0653	CMP sensor and APP sensor 5 V circuit high input		
P0660	Intake shutter solenoid valve (half) control circuit open		—
P0661	Intake shutter solenoid valve (half) control circuit low input		—
P0662	Intake shutter solenoid valve (half) control circuit high input		—
P0663	Intake shutter solenoid valve (full) control circuit open		—
P0664	Intake shutter solenoid valve (full) control circuit low input		—
P0665	Intake shutter solenoid valve (full) control circuit high input		—
P0685	Main relay control circuit open	—	—
P0698	EGR valve position sensor 5 V circuit low input		
P0699	EGR valve position sensor 5 V circuit high input		
P1196	Key off voltage high input	—	—
P1259	IMMOBILIZER to PCM signal error		—
P1260	Theft detected Vehicle immobilizer		—
P1391	Glow plug control circuit low input		—
P1392	Glow plug control circuit high input		—
P1528	Exhaust shutter solenoid valve circuit problem		—
P1602	Immobilizer/PCM communication error		—
P1603	ID Number Unregistered		—
P1604	Code word unregistered		—
P1621	Immobilizer code word does not match		—
P1622	Immobilizer ID does not match		—
P1623	Immobilizer code word/ID number write failure		—
P1624	Anti-theft system		—
P1675	Injection quantity adjustment value writing error		—
P1676	Injection quantity adjustment value checksum error		—
P2008	Variable swirl control (VSC) solenoid valve control circuit open		—
P2009	Variable swirl control (VSC) solenoid valve circuit low input		—
P2010	Variable swirl control (VSC) solenoid valve circuit high input		—
P2135	Accelerator pedal position (APP) sensor No.1/No.2 voltage correlation problem		
P2136	Accelerator pedal position (APP) sensor No.1/idle switch voltage correlation problem		
P2143	EGR control solenoid valve control circuit open		—
P2144	EGR control solenoid valve control circuit low input		
P2145	EGR control solenoid valve control circuit high input		
P2146	Fuel injector No.1 and No.4 power supply circuit malfunction	—	

ON-BOARD DIAGNOSTIC [WL-C, WE-C]

DTC	Condition	KOEO	KOER
P2147	Fuel injector No.1 and No.4 power supply circuit low input	—	
P2148	Fuel injector No.1 and No.4 power supply circuit high input	—	
P2149	Fuel injector No.2 and No.3 power supply circuit malfunction	—	
P2150	Fuel injector No.2 and No.3 power supply circuit low input	—	
P2151	Fuel injector No.2 and No.3 power supply circuit high input	—	
P2227	BARO sensor circuit range/performance malfunction	—	—
P2228	BARO sensor circuit low input		
P2229	BARO sensor circuit high input		
P2263	Air charging pressure performance malfunction	—	—
P2530	PCM (engine switch circuit) malfunction		—